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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2025-2026

SC2006/CZ2006 – SOFTWARE ENGINEERING

Nov/Dec 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer **ALL** questions.
3. This is an open-book examination.
4. All questions carry equal marks.
5. Refer to Appendix A on page 4 for the project description which is needed to answer some of the questions.

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1. Based on the project description given in Appendix A,

   (a) Identify the actors and use cases of the project and draw the use case diagram. Use <<include>> and <<extend>> relationships where appropriate.

   (18 marks)

   (b) Write the use case description for the functionality of the door access by a guest. Your use case description must include the following elements: Actors, Pre-conditions, Flow of events and Alternative flows, if any.

   (7 marks)

2. (a) Draw a state machine diagram for the temporary guest access controller that shows the states and transitions of the controller.

   (4 marks)

Note: Question No. 2 continues on Page 2

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(b) Draw an activity diagram with swim lanes if any, that shows the activities for the process of unlocking the door via the voice assistant by a guest.

(8 marks)

(c) Compare the Waterfall model and Agile methodology in terms of flexibility, customer involvement, and delivery approach. Give one advantage for each model.

(5 marks)

(d) Scrum projects are managed in Sprints. Explain two advantages and two challenges of working in Sprints with examples.

(8 marks)

3. (a) Based on the project description given in Appendix A,

(i) Propose an appropriate architecture for the Smart Home Control System (SHCS), and draw a detailed Class diagram with key attributes and methods in each class to reflect the architecture design.

(8 marks)

(ii) Identify the one relevant non-functional requirement for the system. Propose an alternative architecture design to address this non-functional requirement.

(8 marks)

(b) Answer the following questions related to software design:

(i) Propose a scenario in the Smart Home Control System (SHCS) where more than one design patterns can be applied together, and draw the class diagram to explain the scenario.

(6 marks)

(ii) Critically evaluate: "*Overusing design patterns can make code unnecessarily complex.*" Do you agree? Why or why not?

(3 marks)

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4. (a) When a new user is created in the Smart Home Control System (SHCS), the following information is required:

1. Name: The name of the user, which must be in ASCII format.
2. Phone Number: an 8-digit integer value from 10000000 to 99999999.
3. Access Expiry Date: a date between 2025 and 2030.
4. Type of the User: one of "Administrator", "Owner", "Guest", "One Time User".

(i) Determine the equivalence classes for the above FOUR inputs.

(4 marks)

(ii) Determine the boundaries of the equivalence classes identified in your answers to Q4(a)(i). For each boundary, identify a value on the boundary, a value just below the boundary, and a value just above the boundary.

(4 marks)

(iii) You intend to perform **defensive testing** of the order information. Design a set of test cases to test the FOUR inputs based on the equivalence classes and boundary values identified in your answers to Q4(a)(i) and Q4(a)(ii).

(5 marks)

(b) Testing automation is one of important topics in software engineering. Explain how you can potentially leverage large language models (LLM) to perform testing automatically.

(4 marks)

(c) Automatic software development is the holy grail in software engineering. Identify at least four challenges in the fully automated software development.

(8 marks)

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Appendix A

Smart Home Control System with LLM-Powered Voice Assistance

A technology company is developing a Smart Home Control System (SHCS), accessible through both a mobile app and a voice assistant powered by a large language model (LLM).

Homeowners are the primary users. Through the mobile app or voice assistant, they can control home devices: lights, air conditioning, and door locks. For example, a homeowner can dim the living room lights, adjust the AC temperature, or lock the front door. When using voice commands, the LLM assistant transcribes the request, interprets its meaning, and may ask clarifying questions if the command is ambiguous. Homeowners can also grant or revoke temporary guest access to door locks. If access is given, the system enforces a strict time window and logs all activity.

Guests are secondary users. They may receive temporary access to specific door locks from the homeowner. Guests can then unlock or lock doors either via the mobile app or by speaking commands to the voice assistant, but only within the access period defined by the homeowner. If a guest attempts access outside the time window, the system denies the request and records the event in the log.

Administrators are responsible for user management. They can add or remove accounts, reset credentials, and adjust access rights. Administrators do not control devices but ensure that the system remains secure and that users are properly authenticated.

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.